

Closed graphs and domain density of maximal monotone operators

Proof of Conjecture 00000008842

ziangni-sys

4 October 2026

Operators and maximality

Let E be a real Banach space with continuous dual E^* and $A : E \rightrightarrows E^*$ a set-valued operator. Its graph and domain are

$$G = \{(x, x^*) : x^* \in A(x)\}, \quad D = \{x : \exists x^* \in A(x)\}.$$

Monotonicity means $\langle x^* - y^*, x - y \rangle \geq 0$ for all $(x, x^*), (y, y^*) \in G$. Maximal monotonicity means monotonicity together with the absence of a proper monotone graph extension. Every subset of $E \times E^*$ is the graph of a set-valued operator, by taking its fibers, so maximality is precisely maximality under inclusion among monotone subsets. No closedness is included in this definition. All closures below use the product norm topology.

A general closedness lemma

Let X be a topological space and $C(p, q)$ a reflexive, symmetric relation whose sections $\{p : C(p, q)\}$ are closed for every q . Suppose $G \subseteq X$ is maximal among sets all of whose pairs satisfy C . Then

$$p \in G \iff C(p, q) \text{ for every } q \in G.$$

The forward implication is pairwise compatibility. For the converse, if p satisfies all those relations, $G \cup \{p\}$ is compatible: the pairs in G already are, the new diagonal pair uses reflexivity, and both orders of each mixed pair use symmetry. Maximality forces $p \in G$. Consequently

$$G = \bigcap_{q \in G} \{p : C(p, q)\},$$

an intersection of closed sets, so G is closed.

Apply this lemma to $X = E \times E^*$ and $C((x, x^*), (y, y^*)) \iff \langle x^* - y^*, x - y \rangle \geq 0$. The relation is reflexive, and it is symmetric because negating both differences leaves the pairing unchanged. Each section is closed by continuity of the duality pairing: evaluation is a bounded bilinear map, with $|x^*(x)| \leq \|x^*\| \|x\|$. Hence a maximal monotone graph is closed. The same argument applies to operators $H \rightrightarrows H$ on a real Hilbert space using its continuous inner product.

Density in the closure

For any topological space and any subset D , the canonical inclusion

$$\iota : D \longrightarrow \overline{D}, \quad \iota(x) = x$$

has dense range. Indeed, a relative open neighborhood of $x \in \overline{D}$ is obtained from an ambient open neighborhood of x ; the latter intersects D by the definition of closure. This proves the domain clause, including when the domain is empty. It asserts relative density in $\overline{D}$, not density in the entire ambient space. Combining it with the closedness lemma proves both clauses of the source conjecture.

Formal verification

Lean proves the general closed-compatibility lemma and the actual **DenseRange** of the subtype inclusion. It defines graphs and monotonicity using continuous-dual evaluation and, separately, inner products; the two `conjecture_8842` theorems prove the full conjunction in these settings. They hold already in real normed spaces and inner-product spaces: completeness is unnecessary.